Aidan Mulligan

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EDUCATION

University of Utah – The John and Marcia Price College of Engineering Salt Lake City, UT
Degree: Honors Bachelor of Science in Chemical Engineering May 2028
GPA: 3.83
Achievements: University of Utah President's Scholarship, Phillip A. and Elizabeth Dougall Endowed Scholarship, Dean's list

SKILLS/COURSEWORK

Technical: Python, Java, CAD, 3D printing, Microsoft Office Suite, Microcontrollers
Laboratory: SEM Microscopy, HPLC Purification, UV/Vis Spectroscopy, Protein/Chemical Synthesis
Notable Courses: Process Engineering, Thermodynamics, Statistics, Object-Oriented Programming, Numerical Methods, Organic Chemistry, Fluid Mechanics (planned 8/26), Heat Transfer (planned 8/26), Mass Transfer (planned 1/27)

RELATED EXPERIENCE

Utah Center for Renewables, Efficiency, and Workforce (U-CREW) Salt Lake City, UT
Research Assistant Student January 2026-present

- Conduct energy assessments for Utah industrial facilities by performing on-site measurements and evaluations
- Identify energy-reduction recommendations, proposing measures that can reduce facility energy use by ≥10%
- Have recommended $46,000 in yearly energy savings to assessment clients
- Work supported by Department of Energy (DOE)

STARS Lab, University of Utah Department of Materials Science and Engineering Salt Lake City, UT
Undergraduate Researcher June 2025-September 2025

- Performed literature review for research into high-temperature alloy for rocket engines and other applications
- Studied the microstructural features of high-temperature alloys for aerospace/energy applications

Barrios Lab, University of Utah Department of Biochemistry Salt Lake City, UT
Undergraduate Researcher October 2024-June 2025

- Worked with a team studying the biochemical roles of protein phosphatases in cancer and other diseases
- Performed chemical synthesis and purification to design and test a chemical sensor for SHP2 phosphatase, improving the ease of developing new anti-cancer drugs based on SHP2 phosphatase

Student Researcher June 2023-August 2023

- Characterized unregulated antiviral drugs for feline infectious peritonitis (FIP); findings influenced FDA regulatory changes, making it easier for thousands of cats to access life-saving treatment
- Co-authored a peer-reviewed publication in the American Journal of Veterinary Research based on this research

Metcalf Lab, University of Utah Department of Pharmacology & Toxicology Salt Lake City, UT
Student Researcher June 2022-August 2022

- Investigated the metabolic effects of seizures in a mouse model of a severe epilepsy (Dravet Syndrome)
- Co-authored a peer-reviewed publication which demonstrated potential for future metabolic-based treatments

ADDITIONAL INFORMATION

University of Utah ChemE Car Club Salt Lake City, UT
Team Member August 2024-Present

- Collaborate with peers to design a car, competing in national and regional AIChE-sponsored ChemE Car competitions

Publications:

- Sri Hari, A., Chan, A. M., Scholl, A., **Mulligan, A.**, Camacho, J., Kearns, I. R., Opazo, G. V., Cheminant, J., Musci, T., Goh, M. J., Venosa, A., Moos, P. J., Golkowski, M., & Metcalf, C. S. (2025). Multiomic Analyses Reveal Brainstem Metabolic Changes in a Mouse Model of Dravet Syndrome. *Cells*, *15*(1), 67. https://doi.org/10.3390/cells15010067
- **Mulligan, A. J.**, & Browning, M. E. (2024). Quality assessment and characterization of unregulated antiviral drugs for feline infectious peritonitis: implications for treatment, safety, and efficacy. *American Journal of Veterinary Research*, *85*(3), 1–9. https://doi.org/10.2460/ajvr.23.10.0221

Other Awards:

- National Merit Scholarship, US Presidential Scholar Nominee, Valedictorian, Eagle Scout